

1. 2013Q10

Which of the following combinations best describes the processes involved in respiration?

	<i>Oxygen is required</i>	<i>Carbon dioxide is released</i>
A.	Conversion of pyruvate to acetyl CoA	Glycolysis
B.	Conversion of pyruvate to acetyl CoA	Krebs Cycle
C.	Oxidative phosphorylation	Glycolysis
D.	Oxidative phosphorylation	Krebs Cycle

2. 2014Q14

Which of the following processes produce ATP?

- (1) Glycolysis
 (2) Krebs cycle
 (3) Conversion of pyruvate to lactic acid
- A. (1) and (2) only
 B. (1) and (3) only
 C. (2) and (3) only
 D. (1), (2) and (3)

3. 2015Q6

Which of the following combinations correctly compares the aerobic respiration and anaerobic respiration of muscle cells?

	<i>Aerobic respiration</i>	<i>Anaerobic respiration</i>
A.	occurs only when oxygen is present	occurs only when oxygen is absent
B.	produces more NADH	produces less NADH
C.	glycolysis takes place	no glycolysis
D.	takes place only inside the mitochondria	takes place only in the cytoplasm

4. 2016Q8

Which of the following combinations correctly matches the biochemical reaction with the type of metabolism involved?

	<i>Biochemical reaction</i>	<i>Type of metabolism</i>
A.	conversion of glucose to pyruvate	anabolism
B.	conversion of protein to amino acid	anabolism
C.	conversion of amino acids to urea	catabolism
D.	conversion of glucose to glycogen	catabolism

5. 2016Q26

Which of the following processes in aerobic respiration release NADH?

- (1) glycolysis
 (2) Krebs cycle
 (3) oxidative phosphorylation
- A. (1) and (2) only
 B. (1) and (3) only
 C. (2) and (3) only
 D. (1), (2) and (3)

6. 2017Q7

Which of the following combinations correctly matches the reaction in aerobic respiration with the location where it takes place?

	<i>Reaction in aerobic respiration</i>	<i>Location</i>
A.	regeneration of NAD	cytoplasm
B.	production of carbon dioxide	cytoplasm
C.	conversion of pyruvate to acetyl-CoA	mitochondrion
D.	conversion of triose phosphate to pyruvate	mitochondrion

7. 2018Q11

Which of the following processes takes place at the inner membrane of mitochondria?

- A. glycolysis
 B. conversion of pyruvate to acetyl CoA
 C. Krebs cycle
 D. oxidative phosphorylation

8. 2018Q26

After vigorous exercise, the blood lactic acid concentration of an athlete increases. Which of the following word equations correctly shows the process that leads to the formation of lactic acid?

- A. glucose $\rightarrow$ lactic acid
 B. glucose $\rightarrow$ lactic acid + water
 C. glucose $\rightarrow$ lactic acid + carbon dioxide
 D. glucose + oxygen $\rightarrow$ lactic acid + carbon dioxide

9. 2019Q3

Which of the following process(es) in the respiratory pathways release(s) carbon dioxide?

- (1) Oxidative phosphorylation
 (2) Reactions in the Krebs cycle
 (3) Conversion of glucose to pyruvate
- A. (1) only
 B. (2) only
 C. (1) and (3) only
 D. (2) and (3) only

10. 2020Q4

Which of the following descriptions of anaerobic respiration in muscles is correct?

- A. It involves glycolysis.
 B. It produces lactic acid and carbon dioxide.
 C. It takes place in the matrix of mitochondria.
 D. It takes place on the inner membrane of mitochondria.

11. 2020Q5

Which of the following combinations correctly matches the stages of cellular respiration and the metabolites produced?

	<i>Stage</i>	<i>Metabolite produced</i>
A.	Glycolysis	acetyl-CoA
B.	Glycolysis	ATP
C.	Krebs Cycle	NAD
D.	Krebs Cycle	pyruvate

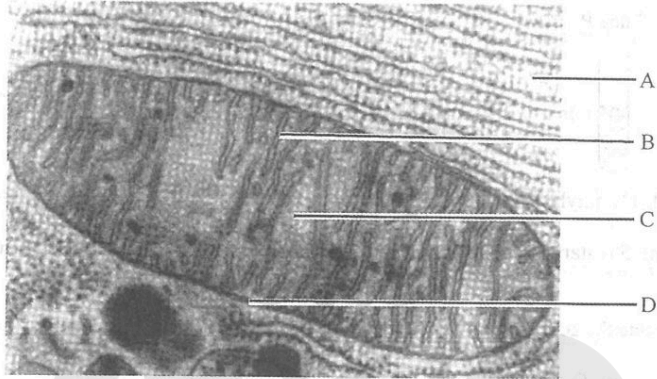
12. 2021Q4

Which of the following pairs of molecules contains the greatest amount of stored energy?

- A. ADP and NAD
- B. ADP and NADH
- C. ATP and NAD
- D. ATP and NADH

13. 2021Q5

Questions 5 and 6 refer to the electron micrograph below, which shows a mitochondrion in an animal cell:



Which of the following substances passes through D into the mitochondrion during aerobic respiration?

- A. glucose
- B. pyruvate
- C. acetyl-CoA
- D. triose phosphate

14. 2021Q6

Anaerobic respiration takes place at

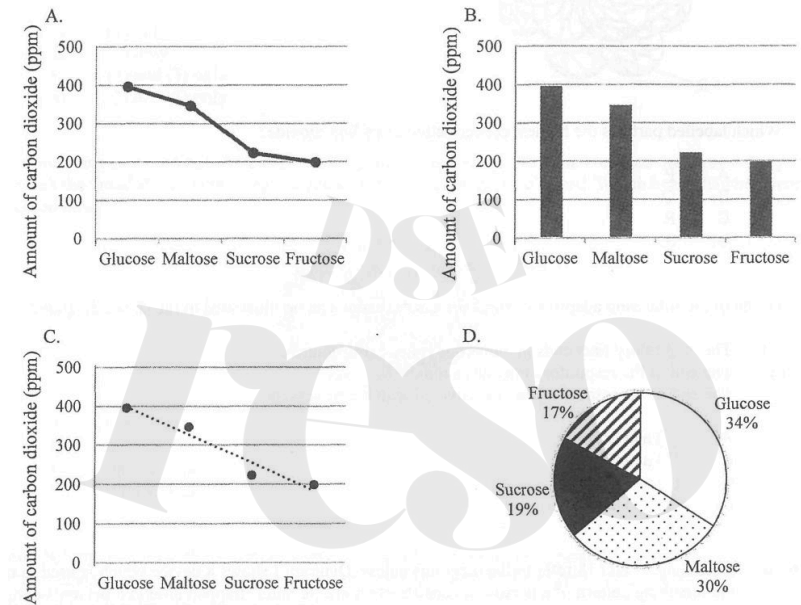
- A. A.
- B. B.
- C. C.
- D. D.

15. 2022Q13

Questions 13 and 14 refer to the table below, which shows the effect of different substrates on the rate of anaerobic respiration of yeast:

Substrate	Average amount of carbon dioxide produced after 10 min (ppm)
Glucose	395.2
Maltose	345.8
Sucrose	222.2
Fructose	198.2

Which of the following graphs is most suitable for presenting the data in the table?



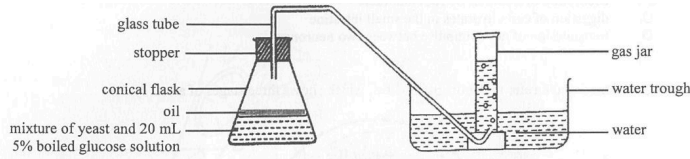
16. 2022Q14

In this investigation, which part of the yeast cell produces carbon dioxide?

- A. cytoplasm
- B. cell membrane
- C. matrix of mitochondria
- D. inner membrane of mitochondria

17. 2023Q7

Questions 7 and 8 refer to the diagram below, which shows a set-up used to investigate the rate of respiration in yeast:



Which of the following word equations correctly shows the reaction being investigated?

- A. glucose $\rightarrow$ carbon dioxide + water
- B. glucose $\rightarrow$ carbon dioxide + ethanol
- C. glucose + oxygen $\rightarrow$ carbon dioxide + water
- D. glucose + oxygen $\rightarrow$ carbon dioxide + ethanol

18. 2023Q8

Which of the following modifications about the set-up can improve the accuracy of the experiment?

- A. use a measuring cylinder instead of a gas jar.
- B. use a larger conical flask to contain the mixture.
- C. use a glass tube with a smaller internal diameter.
- D. use 30 mL instead of 20 mL of 5% boiled glucose solution.

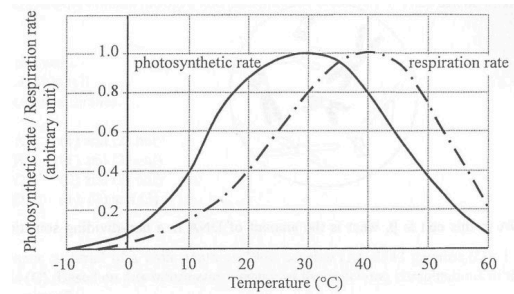
19. 2023Q11

Which of the following processes produces ATP?

- (1) glycolysis
- (2) Calvin cycle
- (3) photochemical reactions
- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

20. 2024Q9

Question 9 refers to the graph below, which shows the changes in the photosynthetic rate and respiration rate of a rice crop at different temperatures while the light intensity and concentration of carbon dioxide were kept constant:



Based on the data shown in the graph, which of the following is / are possible explanation(s) of the drop in the two rates?

- (1) Transpiration rate increases at higher temperatures.
- (2) The enzymes involved denature at higher temperatures.
- (3) The two processes take place at different organelles of the cell.
- A. (1) only
- B. (2) only
- C. (1) and (3) only
- D. (2) and (3) only

21. 2024Q10

Lactic acid is produced during the anaerobic respiration in humans. Which of the following combinations correctly describes the metabolism of the lactic acid produced?

- | | | |
|----|--|----------------------------|
| | the location where lactic acid is broken down | the period involved |
| A. | muscle | during vigorous exercise |
| B. | muscle | after vigorous exercise |
| C. | the liver | during vigorous exercise |
| D. | the liver | after vigorous exercise |

LQ

1. 2012Q9

Drugs X and Y may inhibit enzymes involved in glycolysis, the Krebs cycle, or oxidative phosphorylation. To study the effects of the drugs, some muscle cells were isolated and treated with these two drugs separately in the presence of oxygen. The cellular levels of ATP, NADH, and pyruvate were determined. The results are shown in the table below:

	ATP	NADH	Pyruvate
Control (without treatment)	100%	100%	100%
Drug X	2%	3%	5%
Drug Y	20%	15%	150%

- The data for the control are set as 100% for comparative purpose.
- a) Suggest the key process that is inhibited by drug X. Explain your answer.

(3 marks)
- b) Suggest the key process that is inhibited by drug Y. Explain why there is an accumulation of pyruvate in the muscle cells after treatment with drug Y.

(3 marks)
- c) Instead of incubating in the presence of oxygen, the untreated muscle cells were incubated under anaerobic conditions. Predict the change in the cellular ATP, NADH and lactate levels.

(3 marks)
- d) A student would like to study the enzymes involved in the glycolysis, the Krebs cycle and oxidative phosphorylation separately. Suggest which cellular components he needs to isolate for the investigation.

(3 marks)

2. 2014Q1
- For each of the biomolecules listed in Column 1, select from Column 2 one phrase that matches it. Put the appropriate letter in the space provided.

Column 1		Column 2
NADPH	_____	A. a product of oxidative phosphorylation
pyruvate	_____	B. a product of photochemical reactions
NAD	_____	C. a product of carbon fixation
		D. a product of glycolysis

3. 2018Q9b
- b) What are the major products of oxidative phosphorylation? State their significance.

(4 marks)

4. 2023Q10c
- c) Cassava is a crop that contains a natural toxin. Consuming inadequately cooked cassava may result in cyanide poisoning. Cyanide shuts down oxidative phosphorylation in mitochondria by inhibiting a key enzyme of the process.

i) Name the structure of the mitochondrion where this enzyme is located.

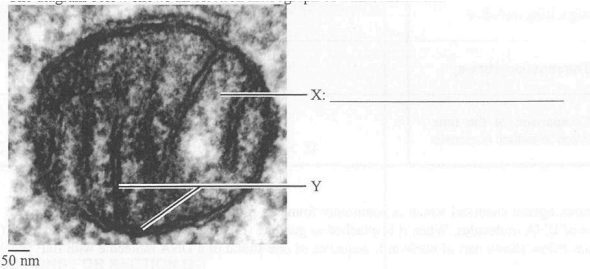
(1 mark)

ii) A man accidentally consumed some raw cassava. How will his blood lactate level change? Explain your answer.

(3 marks)

5. 2024Q3

The diagram below shows an electron micrograph of a mitochondrion:



- a) Label X in the above diagram.

(1 mark)
- b) Describe **one** observable feature of Y and explain how this feature is related to the functioning of mitochondria.

(2 marks)
- c) Chemical Z can inhibit an enzyme found in X.

i) Which key process of respiration would be inhibited?

(1 mark)

ii) If chemical Z is added to a plant cell culture, how would this affect the respiratory pathway?

(3 marks)

Ans

2013Q10: D (67%)	2014Q14: A (45%)	2015Q6: B (53%)	2016Q8: C (64%)
2016Q26: A (65%)	2017Q7: C (58%)	2018Q11: D (69%)	2018Q26: A (50%)
2019Q3: B (61%)	2020Q4: A (64%)	2020Q5: B (73%)	2021Q4: D (85%)
2021Q5: B (42%)	2021Q6: A (62%)	2022Q13: B (69%)	2022Q14: A (55%)
2023Q7: B (68%)	2023Q8: A (67%)	2023Q11: B (54%)	2024Q9: B (71%)
2024Q10: D (38%)			

2012Q9

- 9 (a) • drug X inhibited glycolysis (1) (1)
- as glycolysis is the first step in the respiratory pathway, the inhibition of glycolysis will halt the processes that follow (1), i.e. Krebs cycle and oxidative phosphorylation (1)
- hence, the overall production of pyruvate, ATP and NADH are greatly reduced (1), showing that the whole respiratory pathway was jeopardized (1)
- OR
- Pyruvate is the product of glycolysis (1) (1)
- As the production of pyruvate is greatly reduced after treating with drug X (1) (1)
- Glycolysis was inhibited in this case (1) (1)
- (b) • drug Y inhibited Krebs cycle (1) (1)
- when the respiratory pathway is halted at Krebs cycle, pyruvate will not be metabolised (1) (1)
- but glycolysis still proceeds as usual and produce pyruvate (1), as a result, pyruvate will accumulate (1)
- in anaerobic conditions, muscle cells undergoes anaerobic respiration which produce less ATP (1) and less NADH (1) than that of aerobic respiration (1),(1)
- lactic acid level rises as it is a product of anaerobic respiration (1) (1)
- glycolysis: cytoplasm (1) (1)
- Krebs cycle: mitochondrial matrix (1) (1)
- oxidative phosphorylation: mitochondrial inner membrane (1) (1)

12 marks

2014Q1

- 1 B (1)
- D (1)
- A (1)

3 marks

2018Q9b

- 9 (b) • it regenerates NAD and FAD (1)
- which are important hydrogen carriers in glycolysis / Krebs cycle (1) (4)
- it leads to the formation of ATP (1)
- which is an important energy source to support cellular activities / reactions (1)

2023Q10c

- 10 (c) (i) • inner membrane / cristae (1) (1)
- (ii) • his blood lactate level will increase (1)
- as the oxidative phosphorylation is shut down, his body cells will switch to anaerobic respiration to obtain energy / rely on glycolysis for energy release (1) (3)
- lactate / lactic acid will be produced in anaerobic respiration (1) and be released into the blood

2024Q3

	Concept for mark award	Example	Marks
3(a)	correct labelling	matrix	(1)
3(b)	correct description of the feature	the inner membrane forms infolding / folds / invagination	(1)
	significance of the feature to the functioning of mitochondria	increased surface area to hold more enzymes / electron carriers / membrane proteins for aerobic respiration / oxidative phosphorylation / synthesis of ATP	(1)
3(c)(i)	correct process	inhibition of the conversion of pyruvate to acetyl co-A / inhibition of Krebs cycle	(1)
3(c)(ii)	consequences for the respiratory pathway	Any three of the following points: - glycolysis proceeds as usual to produce pyruvate - pyruvate cannot be metabolised through Krebs cycle / no co-enzyme A is regenerated for the conversion of pyruvate to acetyl co-A - ATP synthesis was inhibited / NADH and FADH₂ formation was inhibited / oxidative phosphorylation was inhibited - pyruvate will be converted to alcohol and carbon dioxide in the anaerobic pathway of plant cells / alcoholic fermentation	(3)

7 marks